

Introduction to Digital Signal Processing

Lecture Notes

Collected Edition, Fall 2026

Lecturer Prof. Justin Haldar

Coverage Lectures 1–5, 7

Updated September 26, 2026

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 1: Signals: Analog, Discrete-Time, and Digital

Lecturer Prof. Justin Haldar

Collected by Gewenxin Yu

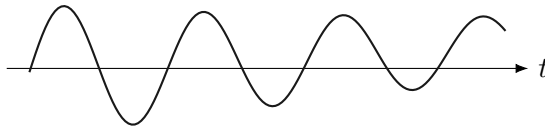
Date August 25, 2026

1 What is a Digital Signal? (1D Case)

We distinguish three flavors of one-dimensional signals, based on how the independent variable (time) and the dependent variable (amplitude) are represented.

1.1 Continuous-Time (“Analog”) Signals

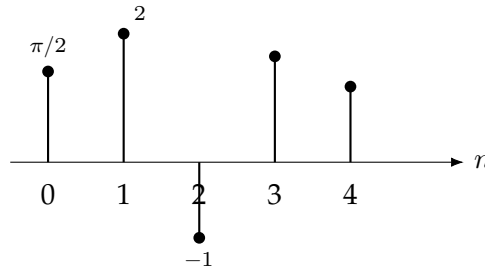
A continuous-time signal $x_a(t)$ is a real- or complex-valued function defined for all real-valued t .



Examples: speech signals, EEG, images formed with focused light.

1.2 Discrete-Time Signals

A discrete-time signal $x[n]$ is defined only for integer-valued n .



Discrete-time signals arise in (at least) two ways:

- **Sampled continuous signals.** $x[n] = x_a(nT)$, e.g. a sampled audio signal.
- **Inherently discrete signals.** Quantities that only update at discrete instants, e.g. the daily maximum temperature, or the number of students late to class on days when class meets.

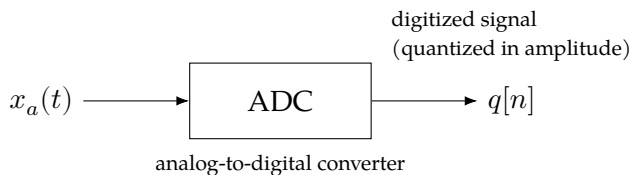
1.3 Digital Signals

Definition (Digital Signal).

A **digital signal** is sampled in time *and* in amplitude: each value $x[n]$ is stored with a finite number of bits, so the numbers are rounded. This is called **finite precision** or **quantization**.

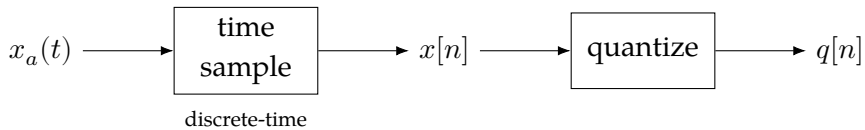
2 From the Real World to Numbers

2.1 The Real ADC



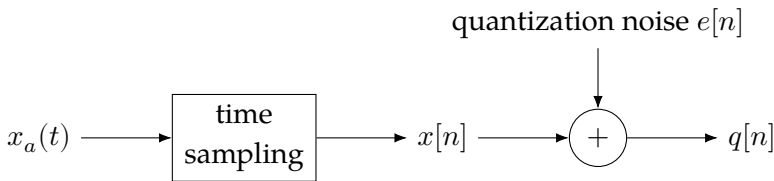
2.2 A Useful Model

We conceptually split the ADC into a time-sampling stage followed by an amplitude-quantization stage:



2.3 An Even More Useful Model

Quantization is modeled as *additive quantization noise* $e[n]$ injected after ideal time-sampling:



Remark.

Discrete-time systems *can* be linear. **Digital systems cannot be linear**; quantization is an inherently nonlinear operation.

3 DSP Topics for the Course

- **Sampling and reconstruction** (ADC and DAC)
 - Nyquist rate
- **Discrete-time linear systems**

$$x_a(t) \rightarrow \boxed{\text{linear system}} \rightarrow y_a(t), \quad x[n] \rightarrow \boxed{\text{linear system}} \rightarrow y[n]$$

- **Signal representation and coding**
 - Discrete Time Fourier Transform (DTFT)
 - Discrete Fourier Transform (DFT)
 - Discrete cosine transform and wavelets (WebP, JPEG, MPEG, MP3, etc.)
- **Implementation**
 - Design of digital filters

-
- Hardware (CPU, DSP, GPU, FPGA)
 - Computational cost (algorithmic) and quantization / overflow (configuration details)

4 Why DSP Instead of Analog Signal Processing?

- Cost
- Storage of data
- Availability
- Easily reconfigured
- Implementation time
- Reliability
- Flexibility

Optional Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 1, 2.1, 2.2, 2.4
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 1.1–1.3

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EE 483: Digital Signal Processing

Lecture 2: Systems: From Continuous-Time to Discrete-Time

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Collected by Gewenxin Yu

Date August 27, 2026

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Last Time / Today

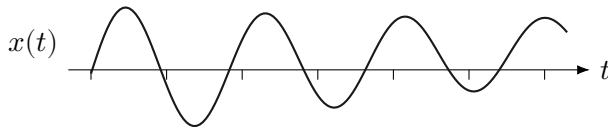
Last time: Defined digital and discrete-time signals.

Today: Systems! Review continuous-time systems, introduce discrete-time systems, and introduce notation and building blocks.

2

1 Review: Continuous-Time Systems

Recall a continuous-time signal (1D) is a real- or complex-valued function defined on the real line.



A **continuous-time system** is a mapping from an input signal $x(t)$ to an output $y(t)$:

$$x(t) \rightarrow \boxed{\begin{array}{c} \text{System} \\ H \end{array}} \rightarrow y(t), \quad y(t) = H\{x(t)\}.$$

Example (Continuous-Time Systems).

$$y(t) = x(t),$$

$$y(t) = x(t - \tau),$$

$$y(t) = \int_0^1 x(t - \tau) d\tau,$$

$$y(t) = |x(t)|.$$

3 1.1 Linear Systems

Definition (Homogeneity).

If $H\{x(t)\} = y(t)$, then $H\{\alpha x(t)\} = \alpha y(t)$ for all $\alpha \in \mathbb{C}$ and all $x(t)$.

Definition (Additivity).

If $H\{x_1(t)\} = y_1(t)$ and $H\{x_2(t)\} = y_2(t)$, then $H\{x_1(t) + x_2(t)\} = y_1(t) + y_2(t)$ for all $x_1(t)$ and $x_2(t)$.

1.2 Time-Invariant Systems

Definition (Time-Invariance).

If $H\{x(t)\} = y(t)$, then $H\{x(t - t_0)\} = y(t - t_0)$ for all $t_0 \in \mathbb{R}$ and all $x(t)$.

4 1.3 Linear Time-Invariant Systems (LTI)

LTI systems are characterized by their **impulse response**:

$$\begin{aligned} h(t) &\triangleq H\{\delta(t)\}, \\ H\{\delta(t - t_0)\} &= h(t - t_0). \end{aligned}$$

Notice that $x(t)$ can be decomposed using the sifting property of the delta

function:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau.$$

Applying $H\{\cdot\}$ to both sides and using linearity + time-invariance:

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad (\textbf{convolution}).$$

5 1.4 Frequency Response

LTI systems are also characterized by their **frequency response** $H(j\Omega)$, the Fourier transform of $h(t)$:

$$\begin{aligned} X(j\Omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt, \quad \Omega \in \mathbb{R}, \\ x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega, \quad t \in \mathbb{R}. \end{aligned}$$

Eigenfunctions of LTI systems. Let $x(t) = e^{j\Omega_0 t}$ with Ω_0 real. Then

$$\begin{aligned} y(t) &= \int h(\tau) x(t - \tau) d\tau \\ &= \int h(\tau) e^{j\Omega_0(t-\tau)} d\tau \\ &= e^{j\Omega_0 t} \int h(\tau) e^{-j\Omega_0 \tau} d\tau \\ &= x(t) H(j\Omega_0). \end{aligned}$$

6 So $x(t) = e^{j\Omega_0 t}$ is an **eigenfunction** of the LTI system, with eigenvalue $H(j\Omega_0)$.
More generally,

$$Y(j\Omega) = H(j\Omega) X(j\Omega).$$

1.5 Transfer Function

LTI systems are also characterized by the **transfer function** $H(s)$, the Laplace transform of $h(t)$:

$$X(s) = \int_{-\infty}^{\infty} e^{-st} x(t) dt, \quad s \in \mathbb{C}, \quad Y(s) = H(s)X(s).$$

- Setting $s = j\Omega$ recovers the Fourier transform.
- Handles more general signals; **regions of convergence (ROC)** tell us about stability and causality.
- Useful for filter design and for solving differential equations.

Example (Solving a Differential Equation with the Laplace Transform).

Consider $\frac{dx}{dt} = -x(t)$. Taking the Laplace transform,

$$sX(s) - x(0) = -X(s) \implies X(s) = \frac{x(0)}{1+s} \implies x(t) = x(0) e^{-t} u(t).$$

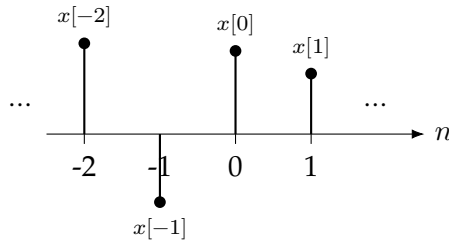
7 2 LTI Discrete-Time Systems (Preview)

Discrete-time LTI systems are characterized analogously:

- Impulse response $h[n]$
- Frequency response $H(e^{j\omega})$: the DTFT
- Transfer function $H(z)$: the z -transform
- Linear (constant-coefficient) difference equations

3 Discrete-Time Signals

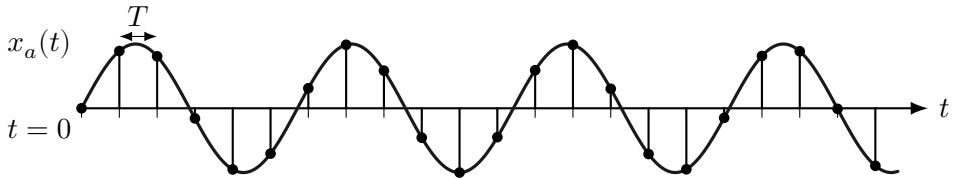
- Represented as a sequence of numbers (“samples”).
- Denoted $x[n]$, where n is an integer, $-\infty < n < \infty$.
- $x[n]$ is **not defined** for non-integer n .



Notation: $\{\dots, x[-2], x[-1], x[0], x[1], \dots\}$, with $n = 0$ at $x[0]$.

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3.1 Sampling of Continuous-Time Signals



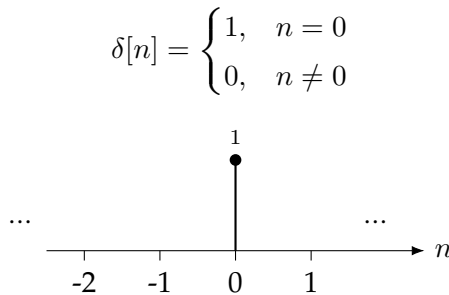
$$x[n] = x_a(t)|_{t=nT} = x_a(nT).$$

T : sampling interval / sampling period.

$f_T = \frac{1}{T}$: sampling rate / sampling frequency.

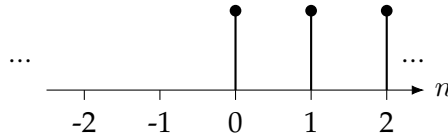
3.2 Basic Sequences

Unit impulse sequence



Unit step sequence

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



9 3.3 Real Sinusoidal Sequences

$$x[n] = A \cos(\omega_0 n + \phi),$$

where A is the amplitude,
 ω_0 is the frequency, and ϕ is the phase.

3.4 Complex (Damped) Sinusoids

$$x[n] = A\alpha^n, \quad -\infty < n < \infty.$$

A and α can be real or complex.

Write $\alpha = e^{\sigma_0} e^{j\omega_0}$ (σ_0, ω_0 real) and $A = |A| e^{j\phi}$. Then

$$x[n] = |A| e^{j\phi} e^{(\sigma_0 + j\omega_0)n} = x_{re}[n] + j x_{im}[n],$$

$$x_{re}[n] = |A| e^{\sigma_0 n} \cos(\omega_0 n + \phi),$$

$$x_{im}[n] = |A| e^{\sigma_0 n} \sin(\omega_0 n + \phi).$$

- $\sigma_0 = 0 \Rightarrow$ constant amplitude
- $\sigma_0 < 0 \Rightarrow$ exponential decay
- $\sigma_0 > 0 \Rightarrow$ exponential growth

10 3.5 Arbitrary Sequences can be represented as a weighted sum of time-shifted versions of basic sequences

Example.

$x[n] = \{0.5, 1, -1, 0.5\}$ with $n = 0$ at the sample 1 can be written as

$$x[n] = 0.5 \delta[n + 1] + \delta[n] - \delta[n - 1] + 0.5 \delta[n - 2],$$

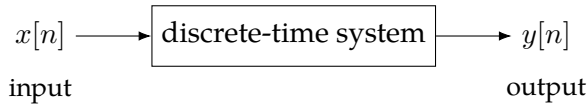
or, in terms of shifted unit steps,

$$x[n] = 0.5 u[n + 1] + 0.5 u[n] - 2 u[n - 1] + 1.5 u[n - 2] - 0.5 u[n - 3].$$

Fact.

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n - m], \quad \text{e.g.} \quad u[n] = \sum_{m=0}^{\infty} \delta[n - m].$$

4 Discrete-Time Systems



11 4.1 Basic Operations

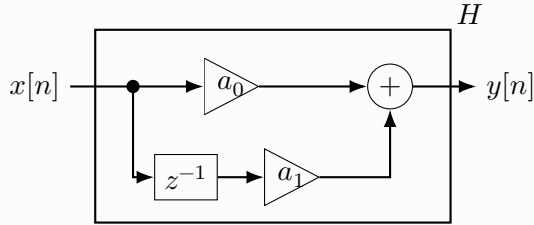
- **Multiplication (scalar):** $x[n] \rightarrow \boxed{A} \rightarrow y[n]$, $y[n] = A x[n]$.
- **Delay:** $x[n] \rightarrow \boxed{z^{-1}} \rightarrow y[n]$, $y[n] = x[n - 1]$.
- **Time reversal:** $x[n] \rightarrow \boxed{\text{time reversal}} \rightarrow y[n]$, $y[n] = x[-n]$.
- **Advance:** $x[n] \rightarrow \boxed{z} \rightarrow y[n]$, $y[n] = x[n + 1]$.
- **Branching:** $x[n] \rightarrow \bullet \rightarrow \begin{matrix} x[n] \\ x[n] \end{matrix}$ $x[n]$ splits into two identical copies.

- **Addition:** $x[n] \rightarrow \bigoplus \rightarrow y[n] = x[n] + w[n]$

- **Product (modulation):** $x[n] \longrightarrow \bigcirc \times \longrightarrow y[n] = x[n]w[n]$
 $\uparrow$
 $w[n]$

Example.

A two-tap system:



$$y[n] = a_0 x[n] + a_1 x[n - 1].$$

Its impulse response is

$$h[n] = H\{\delta[n]\} = a_0 \delta[n] + a_1 \delta[n - 1] = \{a_0, a_1\}, \quad n = 0 \text{ at } a_0.$$

This is an **FIR filter** (Finite Impulse Response).

Example (Moving-Average System).

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n - k].$$

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 4.1–4.4
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 1.4

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Lecture 3: Properties of Discrete-Time Systems

Lecturer Prof. Justin Haldar

Collected by Gewenxin Yu

Date September 1, 2026

1

Last Time

Systems building blocks.

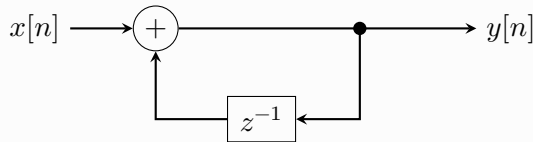
Example (FIR Filter).

A finite impulse response system, as introduced last lecture.

Example (Accumulator System).

$$y[n] = \sum_{\ell=-\infty}^n x[\ell], \quad y[n-1] = \sum_{\ell=-\infty}^{n-1} x[\ell] \implies y[n] - y[n-1] = x[n].$$

This can be calculated practically as $y[n] = y[n-1] + x[n]$ (given some initial condition):



The impulse response is $h[n] = H\{\delta[n]\} = u[n]$. This system is **not FIR**; it is **IIR** (Infinite Impulse Response).

Properties we will study: [Linearity](#), [Time Invariance](#), [Causality](#), [Impulse Response](#), [Convolution](#), [Stability](#).

2 1 Linear Discrete-Time Systems

Definition (Linearity).

For arbitrary $\alpha, \beta \in \mathbb{C}$ and arbitrary inputs $x_1[n]$ and $x_2[n]$: if

$$y_1[n] = H\{x_1[n]\} \quad \left(x_1[n] \xrightarrow{H} y_1[n] \right)$$

and if

$$x_2[n] \xrightarrow{H} y_2[n],$$

then if

$$x[n] = \alpha x_1[n] + \beta x_2[n],$$

we must have

$$x[n] \xrightarrow{H} y[n] = \alpha y_1[n] + \beta y_2[n].$$

Example (Accumulator is Linear).

$$y[n] = \sum_{\ell=-\infty}^n x[\ell] \quad (\text{Accumulator})$$

$$x_1[n] \xrightarrow{H} y_1[n] = \sum_{\ell=-\infty}^n x_1[\ell]$$

$$x_2[n] \xrightarrow{H} y_2[n] = \sum_{\ell=-\infty}^n x_2[\ell]$$

If

$$x[n] = \alpha x_1[n] + \beta x_2[n],$$

then

$$\begin{aligned} x[n] \xrightarrow{H} y[n] &= \sum_{\ell=-\infty}^n (\alpha x_1[\ell] + \beta x_2[\ell]) \\ &= \alpha \sum_{\ell=-\infty}^n x_1[\ell] + \beta \sum_{\ell=-\infty}^n x_2[\ell] \\ &= \alpha y_1[n] + \beta y_2[n]. \end{aligned}$$

Therefore, the accumulator is linear.

3 2 Homogeneity for the Accumulator, via the Difference-Equation Form

Example.

$$y[n] = y[n - 1] + x[n] \quad (\text{accumulator})$$

Assume

$$x[n] \xrightarrow{H} y[n]$$

Let

$$x_1[n] = \alpha x[n] \xrightarrow{H} y_1[n]$$

Then

$$y_1[n] = y_1[n - 1] + \alpha x[n], \quad \alpha y[n] = \alpha y[n - 1] + \alpha x[n].$$

Subtracting,

$$\begin{aligned} y_1[n] - \alpha y[n] &= y_1[n - 1] - \alpha y[n - 1] \\ \Rightarrow g[n] &= g[n - 1], \quad g[n] \triangleq y_1[n] - \alpha y[n]. \end{aligned}$$

Remark.

$g[n]$ should be zero for all n for homogeneity to hold.

We assume **zero initial conditions (ICs)** (a restrictive definition; we will use something more general for LTI systems):

$$\begin{aligned} \exists M \in \mathbb{Z} \text{ s.t. } y[n] &= 0, \quad x[n] = 0, \quad y_1[n] = 0 \quad \forall n \leq M \\ \Rightarrow g[n] &= 0 \quad \forall n \leq M \\ \Rightarrow g[n] &= 0 \quad \forall n. \end{aligned}$$

So $y_1[n] = \alpha y[n]$ for all n : **Homogeneity** ✓.

Remark.

Proving linearity is frequently harder than disproving linearity.

Example (Median Filter: Homogeneous but not Additive).

$$y[n] = \text{median}\{x[n-1], x[n], x[n+1]\}.$$

- Homogeneity? ✓
- Additivity? ✗

For single impulses $x_1[n]$ and $x_2[n]$ at different locations, the median filter outputs $y_1[n]$ and $y_2[n]$ that are each zero (or small); but for $x_1[n] + x_2[n]$ (two nonzero samples), the median output $y[n] \neq y_1[n] + y_2[n]$. Hence **additivity fails**.

5 3 Time-Invariant Discrete-Time Systems

Definition (Time-Invariance).

For arbitrary $x[n] \xrightarrow{H} y[n]$: if $x_1[n] = x[n - n_0]$ for arbitrary $n_0 \in \mathbb{Z}$, then $x_1[n] \xrightarrow{H} y_1[n] = y[n - n_0]$.

Example (FIR Filter is Time-Invariant).

$$y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] \quad (\text{FIR filter}).$$

Let $x_1[n] = x[n - n_0]$. Then

$$y_1[n] = \sum_{k=0}^{M-1} \alpha_k x_1[n-k] = \sum_{k=0}^{M-1} \alpha_k x[(n-k)-n_0] \stackrel{?}{=} y[n-n_0] \implies \text{time invariant!}$$

Example (A System that is Not Time-Invariant).

$$y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] + x[0].$$

For $x[n] = \delta[n]$:

$$y[n] = \begin{cases} \alpha_n + 1, & 0 \leq n \leq M-1 \\ 0 + 1, & \text{else} \end{cases}.$$

For $x_1[n] = \delta[n-1]$:

$$y_1[n] = \begin{cases} \alpha_{n-1} + 0, & 1 \leq n \leq M \\ 0 + 0, & \text{else} \end{cases}.$$

Since $y_1[n] \neq y[n-1]$ (the added constant $x[0]$ does not shift), **time invariant?**
Nope.

The extra term always reads the input at index 0, not at a lag relative to n . Shifting the input by n_0 therefore does not translate that contribution through time; the output is not a shifted copy of the original response.

4 Linear Time-Invariant (LTI) Systems

- Linear
- Time-invariant
- Easy to analyze, characterize, implement, and design.

4.1 Important Subclass: Linear Constant-Coefficient Difference Equations (LDEs)

An important subclass, with zero ICs:

$$\sum_{k=0}^{N-1} d_k y[n-k] = \sum_{q=0}^{M-1} p_q x[n-q].$$

Solving for $y[n]$:

$$y[n] = \sum_{k=1}^{N-1} \tilde{d}_k y[n-k] + \sum_{q=0}^{M-1} \tilde{p}_q x[n-q],$$

$$\left(\tilde{d}_k = -\frac{d_k}{d_0}, \quad \tilde{p}_q = \frac{p_q}{d_0}; \text{ assuming } d_0 \neq 0 \right).$$

7

Examples: Accumulator (FIR Filter).

$$y[n] = y[n-1] - x[n-1] + \rho(y[n-1] - x[n-1]),$$

where $y[n]$ is the loan balance, $y[n-1]$ the previous balance, $x[n-1]$ the payment, and ρ the interest rate. Similar in spirit to the continuous-time ODE form

$$y(t) = \sum_{n=0}^N \alpha_n \frac{d^n x(t)}{dt^n} + \sum_{m=1}^M \beta_m \frac{d^m y(t)}{dt^m}.$$

5 Causal Systems

Definition (Causality).

The m th output sample $y[m]$ depends only on input samples $x[n]$ with $n \leq m$. This must be true for all possible inputs and all m .

Examples.

- $y[n] = \text{median}\{x[n-1], x[n], x[n+1]\}$: **not causal**.
- $y[n] = y[n-1] + x[n]$: causal, with an IC.
- $y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k]$: **causal**.
- $y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] + x[0]$: **not causal**.

8

Property (Causal Systems and Matching Inputs).

For a causal system, if $x_1[n] \xrightarrow{H} y_1[n]$ and $x_2[n] \xrightarrow{H} y_2[n]$, and if $x_1[n] = x_2[n]$ for all $n < N$, and the ICs are the same, then $y_1[n] = y_2[n]$ for all $n < N$.

6 Stability

Definition (BIBO Stability).

A system is **bounded-input bounded-output (BIBO) stable** if every bounded input yields a bounded output, i.e., if $|x[n]| \leq B_x < \infty$ for all n , then $|y[n]| \leq B_y < \infty$ for all n .

9

Example (Moving Average is BIBO Stable).

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k].$$

Assume

$$|x[n]| \leq B_x < \infty \quad \text{for all } n.$$

Then

$$\begin{aligned} |y[n]| &= \left| \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] \right| \\ &\leq \frac{1}{M} \sum_{k=0}^{M-1} |x[n-k]| \\ &\leq \frac{1}{M} \sum_{k=0}^{M-1} B_x = B_x \implies \text{BIBO stable!} \end{aligned}$$

Example (Accumulator is Not BIBO Stable).

$y[n] = y[n-1] + x[n]$, with IC $y[n] = x[n] = 0$ for $n < 0$. Let $x[n] = u[n]$ (bounded). Then $y[n] = (n+1)u[n]$, which is **not bounded**.

7 Impulse Response

Definition (Impulse Response).

The impulse response $h[n]$ is the output of a discrete-time system to the input $\delta[n]$.

Example (Accumulator).

$$y[n] = \sum_{k=-\infty}^n x[k] \implies h[n] = \sum_{k=-\infty}^n \delta[k] = u[n].$$

10

Fact.

LTI systems with zero ICs are **completely characterized** by their impulse responses.

Remark.

A more general definition of zero ICs for LTI systems: if $x[n] = 0$ for all n , then $y[n] = H\{x[n]\} = 0$ for all n : “zero-input” $\Rightarrow$ “zero-output.” (Non-zero ICs $\Rightarrow$ not LTI.)

Proof of fact: next time.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 4.5–4.7
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 1.5, 2.4

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Lecture 4: Convolution, System Properties, and Difference Equations

Lecturer Prof. Justin Haldar*Collected by* Gewenxin Yu*Date* September 3, 2026

1 Impulse Response and the Convolution Sum

$$h[n] = H\{\delta[n]\}. \quad (\text{Mitra §4.3})$$

Fact.

LTI systems with zero initial conditions (ICs) are **completely characterized** by $h[n]$.

Proof of fact. Write

$$x[n] = \sum_k x[k] \delta[n - k]^1 \quad (\text{Mitra 4.15})$$

$y[n] = H\{x[n]\}$. If $H\{\cdot\}$ is LTI, then

$$\begin{aligned} y[n] &= H\left\{\sum_k x[k] \delta[n - k]\right\} \\ &= \sum_k x[k] H\{\delta[n - k]\} \quad (\text{linearity}) \\ &= \sum_k x[k] h[n - k]. \quad (\text{time-invariance}) \quad \blacksquare \quad (\text{Mitra 2.20a, 4.16a}) \end{aligned}$$

The summation

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \quad (\text{Mitra 2.20a, 4.16a})$$

is called the **convolution sum** of the sequences $x[n]$ and $h[n]$:

$$y[n] = x[n] \circledast h[n]. \quad (\text{Mitra 2.21})$$

2

Example (Computing the Convolution Sum).

We compute $y[n] = x[n] \circledast h[n]$ numerically for a short input and a short impulse response. Here $x[n]$ is the input and $h[n] = H\{\delta[n]\}$ is the impulse response of an LTI system with zero ICs. By definition,

$$y[n] = \sum_{m=-\infty}^{\infty} x[m] h[n-m]. \quad (\text{Mitra 2.20a, 4.16a})$$

Equivalently, flip $h[n]$ to form $g[m] \triangleq h[-m]$, then slide $g[m]$ relative to $x[m]$: at each n , multiply the overlapping samples and add.

Let $h[n] = \{1, 1, 1\}$ and $x[n] = \{1, 2, 3\}$, with $n = 0$ at the first listed sample of each sequence.

Flip $h[n]$ to $g[m] = h[-m]$; align $x[m]$ and $g[m]$. At $n = 0$:

m	-1	0	1	
$x[m]$	1	2	3	
$g[m]$	1	1	0	
$x[m]g[m]$	1	2	0	$\Rightarrow y[0] = 3$

Shift $g[m]$ one sample to the right to compute $y[1] = 6$, and continue for each n . Then

$$y[0] = \sum_{m=-\infty}^{\infty} x[m] h[0-m] = \sum_{m=-\infty}^{\infty} x[m] g[m], \quad g[m] \triangleq h[-m].$$

¹Following Mitra (§4.3): in $\sum_k x[k] \delta[n-k]$, the index k labels the k th sample $x[k]$ of $\{x[n]\}$. In Eqs. (4.16a)–(4.16b), k is the convolution summation index; Eq. (4.16b) forms $y[n]$ as a sum of input samples $x[n-k]$ weighted by $h[k]$ (see Mitra, Fig. 4.7).

2

With

$$g[m] = \{\dots, 0, 1, 1, 1, 0, 0, \dots\}, \quad x[m] = \{\dots, 0, 1, 2, 3, 0, \dots\},$$

with $m = 0$ at the samples 1 and 2, respectively.

$$y[0] = 0 + 1 + 2 + 0 + 0 = 3, \quad y[1] = \sum_{m=-\infty}^{\infty} x[m] g[m-1] = 6.$$

Continuing in this way,

$$y[n] = \{\dots, 0, 1, 3, 6, 5, 3, 0, \dots\}, \quad n = 0 \text{ at the sample 3.}$$

3

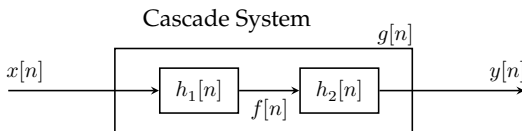
1.1 Convolution Properties

Commutative: $x[n] \otimes h[n] = h[n] \otimes x[n].$ (Mittra 4.17)

Associative: $(x[n] \otimes h[n]) \otimes y[n] = x[n] \otimes (h[n] \otimes y[n]).$ (Mittra 4.18)

Distributive: $x[n] \otimes (h[n] + y[n]) = x[n] \otimes h[n] + x[n] \otimes y[n].$ (Mittra 4.19)

1.2 Cascade System

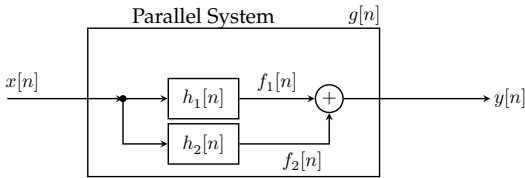


$$\begin{aligned} f[n] &= x[n] \otimes h_1[n], \\ y[n] &= f[n] \otimes h_2[n] \\ &= (x[n] \otimes h_1[n]) \otimes h_2[n] \end{aligned}$$

$$y[n] = x[n] \otimes g[n], \quad g[n] = h_1[n] \otimes h_2[n].$$

(Mittra 4.28)

4 1.3 Parallel System



$$y[n] = f_1[n] + f_2[n],$$

$$f_1[n] = x[n] \otimes h_1[n],$$

$$f_2[n] = x[n] \otimes h_2[n],$$

$$y[n] = x[n] \otimes h_1[n] + x[n] \otimes h_2[n]$$

$$y[n] = x[n] \otimes g[n], \quad g[n] = h_1[n] + h_2[n].$$

(Mitra 4.31)

5 1.4 $h[n]$ and Causality

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad (\text{Mitra 2.20a, 4.16a})$$

$$= \sum_{k=-\infty}^{\infty} x[n-k] h[k]. \quad (\text{Mitra 2.20b, 4.16b})$$

Causality.

$y[n]$ is independent of $x[m]$ for all $m > n \iff h[n] = 0$ for all $n < 0$.² An LTI discrete-time system is **causal** if and only if its impulse response $h[n]$ is a “causal” sequence, i.e., $h[n] = 0$ for all $n < 0$ (Mitra 4.27).

²The handout uses $\Rightarrow$; here we use $\Leftrightarrow$ since both directions hold for LTI systems.

Example.

The FIR system below uses only present and past inputs ($k \geq 0$):

$$\begin{aligned}
 y[n] &= \sum_{k=0}^M \alpha_k x[n-k] \\
 \Rightarrow h[n] &= \sum_{k=0}^M \alpha_k \delta[n-k] \\
 &= \begin{cases} \alpha_n, & 0 \leq n \leq M \\ 0, & \text{else} \end{cases}
 \end{aligned}$$

Since $h[n] = 0$ for all $n < 0$, the system is causal: $y[n]$ depends only on $x[m]$ with $m \leq n$. (Mitra 4.14)

6

1.5 $h[n]$ and Stability**BIBO Stability**

3.

An LTI discrete-time system is **BIBO stable** if and only if its impulse response $\{h[n]\}$ is **absolutely summable**, i.e.,

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty. \quad (\text{Mitra 4.20, 2.42})$$

Proof (sufficiency). Assume $\{x[n]\}$ is bounded, $|x[n]| \leq B_x < \infty$ for all n , where B_x is a finite bound on the input magnitude; then the output amplitude at

³cf. Mitra Sec. 4.4.3.

time instant n , from Eq. (4.16b), is

$$\begin{aligned}
 |y[n]| &= \left| \sum_{k=-\infty}^{\infty} h[k] x[n-k] \right| \\
 &\leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]|^4 \\
 &\leq B_x \sum_{k=-\infty}^{\infty} |h[k]| \\
 &= B_x S < \infty.
 \end{aligned}
 \tag{Mitra 2.20b, 4.16b, 4.21}$$

So if

$$S = \sum_{k=-\infty}^{\infty} |h[k]| < \infty, \tag{Mitra 4.20, 2.42}$$

then $|y[n]| \leq B_y = B_x S < \infty$ for all n ; hence absolute summability is a **sufficient** condition for BIBO stability. ■

7 Necessity. Let $x[n] = \text{sgn}(h[-n])$, (Mitra 4.22) where for real-valued α ,

$$\text{sgn}(\alpha) = \begin{cases} +1, & \alpha > 0 \\ -1, & \alpha < 0 \\ 0, & \alpha = 0. \end{cases}$$

Then

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k] = \sum_{k=-\infty}^{\infty} h[k] \text{sgn}(h[k-n]).$$

At $n = 0$,

$$y[0] = \sum_{k=-\infty}^{\infty} h[k] \text{sgn}(h[k]) = \sum_{k=-\infty}^{\infty} |h[k]| = S. \tag{Mitra 4.23}$$

⁴The summation $\sum_k$ adds all terms $h[k]x[n-k]$. By the triangle inequality, the absolute value of a sum is at most the sum of absolute values, so $|\sum_k h[k]x[n-k]| \leq \sum_k |h[k]x[n-k]|$.

So for $y[0]$ to be finite (which is necessary for $y[n]$ to be bounded), it is **necessary** that $\sum_{k=-\infty}^{\infty} |h[k]| < \infty$. ■

8

Example (Mitra Ex. 4.16).

Consider a causal LTI discrete-time system with an impulse response given by $h[n] = \alpha^n u[n]$. For this system,⁴

$$\begin{aligned}
 S &= \sum_{n=-\infty}^{\infty} |h[n]| \\
 &= \sum_{n=-\infty}^{\infty} |\alpha^n u[n]| \\
 &= \sum_{n=-\infty}^{\infty} |\alpha^n| u[n] \\
 &= \sum_{n=0}^{\infty} |\alpha|^n \\
 &= \frac{1}{1 - |\alpha|} \quad \text{for } |\alpha| < 1. \quad ^5
 \end{aligned}$$

Therefore, $S < \infty$ if $|\alpha| < 1$, for which the above system is BIBO stable. On the other hand, if $|\alpha| \geq 1$, the infinite series $\sum_{n=0}^{\infty} |\alpha|^n$ does not converge, and the above causal system is not BIBO stable.

⁴Since $|ab| = |a||b|$ and $u[n] \in \{0, 1\}$, we have $|u[n]| = u[n]$ and thus $|\alpha^n u[n]| = |\alpha^n| u[n]$.

⁵The last step applies the geometric series formula, which gives a finite sum when the common ratio has magnitude less than one.

Important Facts.

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad |a| < 1 \quad (\text{Mitra Ex. 4.6})$$

$$\sum_{n=0}^{N-1} a^n = \begin{cases} \frac{1-a^N}{1-a}, & a \neq 1 \\ N, & a = 1. \end{cases}$$

9 2 Linear Difference Equations

$$\sum_{k=0}^N d_k y[n-k] = \sum_{q=0}^M p_q x[n-q], \quad (\text{Mitra 4.32})$$

finite order $\triangleq \max(N, M)$.

If we assume causality, then $y[n]$ can be computed for all $n \geq n_0$ from $x[n]$ for $n \geq n_0$ and the ICs

$$y[n_0-1], y[n_0-2], \dots, y[n_0-N], \quad x[n_0-1], x[n_0-2], \dots, x[n_0-M]. \quad (\text{Mitra 4.33})$$

$$y[n] = - \sum_{k=1}^N \frac{d_k}{d_0} y[n-k] + \sum_{q=0}^M \frac{p_q}{d_0} x[n-q]. \quad (\text{Mitra 4.33})$$

- **FIR** (finite impulse response): $N = 0, M \geq 0$.
- **IIR** (infinite impulse response): $N > 0, M \geq 0$.

10 2.1 FIR Systems

$$y[n] = \sum_{q=0}^M \tilde{p}_q x[n-q] \implies h[n] = \begin{cases} \tilde{p}_n, & n = 0, \dots, M \\ 0, & \text{else.} \end{cases} \quad (\text{Mitra 4.14})$$

Generically called “moving average” filters:

- Moving average: $y[n] = \frac{x[n] + x[n-1]}{2}$. (Mitra 4.4)
- Finite difference: $y[n] = x[n] - x[n-1]$.

- Simple echo: $y[n] = x[n] + \alpha x[n - M]$.

2.2 IIR Systems

$$y[n] = \sum_{k=1}^N \tilde{d}_k y[n - k] + \sum_{q=0}^M \tilde{p}_q x[n - q]. \quad (\text{Mitra 4.33})?$$

- $M = 0$: **autoregressive (AR)**.
- $M > 0$: **autoregressive moving average (ARMA)**.
- Recursive (feedback loop); harder to characterize:
 - Requires an IC (infinitely many solutions for $y[n]$ given only $x[n]$ with unspecified ICs).
 - Harder in general to compute the impulse response.

11 2.3 Example: $y[n] = a y[n - 1] + x[n]$

What is the impulse response?

1. Zero ICs ($y[n] = 0$ for all $n < 0$).

$$\begin{aligned} y[0] &= a y[-1] + \delta[0] = 1, \\ y[1] &= a y[0] + \delta[1] = a, \\ y[2] &= a y[1] + \delta[2] = a^2, \\ &\vdots \\ y[n] &= a^n, \quad n \geq 0 \implies h[n] = a^n u[n]. \end{aligned} \quad (\text{Mitra Ex. 4.16})$$

2. $y[n] = 0$ for all $n \geq 0$. Note $y[n] = a y[n-1] + x[n] \Rightarrow y[n-1] = a^{-1}(y[n] - x[n])$, so

$$\begin{aligned} y[-1] &= a^{-1}(y[0] - \delta[0]) = -a^{-1}, \\ y[-2] &= a^{-1}(y[-1] - \delta[-1]) = -a^{-2}, \\ &\vdots \\ y[n] &= -a^n, \quad n \leq -1 \implies h[n] = -a^n u[-1 - n]. \end{aligned}$$

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2.4 Why?

If we have a particular solution $y_p[n]$ that satisfies a given LDE, then any solution $y_h[n]$ to the “homogeneous LDE” ($x[n] = 0$) gives another valid solution of the form

$$y[n] = y_p[n] + y_h[n], \quad (\text{Mitra 4.39})$$

$$\text{Homogeneous LDE: } \sum_{k=0}^N d_k y[n-k] = 0. \quad (\text{Mitra 4.40})$$

Homogeneous LDE.

$y[n] = a y[n-1] \Rightarrow$ no unique solution! $y[n] = A a^n$ is a solution for any A .
Particular solutions:

$$(1) \quad h[n] = a^n u[n],$$

$$(2) \quad h[n] = -a^n u[-1-n].$$

General solution: $h[n] = a^n u[n] + A a^n$.

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Using $u[-1-n] = 1 - u[n]$,

$$-a^n(1 - u[n]) = a^n u[n] - a^n.$$

Consider candidate solutions to a generic homogeneous equation $\sum_{k=0}^N a_k y[n-k] = 0$ of the form $y[n] = A_\ell \lambda_\ell^n$. For what choices of A_ℓ and λ_ℓ is this a valid solution?

$$\sum_{k=0}^N a_k A_\ell \lambda_\ell^{(n-k)} = 0$$

$$A_\ell \lambda_\ell^n \sum_{k=0}^N a_k \lambda_\ell^{-k} = 0$$

Note: $\sum_{k=0}^N a_k z^{-k}$ is a polynomial in z^{-1} . In the case of simple roots, the homo-

geneous equation has solutions

$$y[n] = \sum_{\ell=1}^N A_{\ell} \lambda_{\ell}^n, \quad (\text{Mitra 4.43})$$

where λ_{ℓ} are the roots of the characteristic polynomial (Mitra §4.6.1) and A_{ℓ} are arbitrary for $\ell = 1, \dots, N \Rightarrow$ infinitely many solutions.

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2.5 Inverse Filter

$$x[n] \rightarrow \boxed{h_1[n]} \rightarrow y[n] \rightarrow \boxed{h_2[n]} \rightarrow x[n]$$

Examples: $h_1[n]$ channel, $h_2[n]$ equalizer; $h_1[n]$ is blur, $h_2[n]$ is deblurring. In either case,

$$h_1[n] \otimes h_2[n] = \delta[n]. \quad (\text{Mitra 4.29})$$

Example.

$y[n] = a y[n-1] + x[n]$ (IIR), with $h_1[n] = a^n u[n]$. Then

$$x[n] = y[n] - a y[n-1] \Rightarrow \text{FIR}, \quad h_2[n] = \delta[n] - a \delta[n-1] \quad (\text{inverse filter}).$$

Caution: ICs should be treated carefully — the inverse filter may not be causal or stable.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 3.2, 3.3
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 2.5, 2.6

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 5: The Discrete-Time Fourier Transform (DTFT)

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1 The Discrete-Time Fourier Transform (DTFT)

Def.

The DTFT $X(e^{j\omega})$ of a sequence $x[n]$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}.^6 \quad (\text{Mitra 3.10})$$

$\Rightarrow$ a complex-valued function of a real-valued variable ω .

$$X(e^{j\omega}) = X_{\text{re}}(e^{j\omega}) + jX_{\text{im}}(e^{j\omega}).$$

Both $X_{\text{re}}(e^{j\omega})$ and $X_{\text{im}}(e^{j\omega})$ are real-valued functions of ω .

$$X(e^{j\omega}) = \underbrace{|X(e^{j\omega})|}_{\text{magnitude}} e^{j\theta(\omega)}, \quad \theta(\omega) = \angle X(e^{j\omega}) \text{ (phase)}.$$

⁶Mitra: Equation (3.10) usually is referred to as the **analysis equation**, because it analyzes how much of each complex exponential signal is present in the original signal.

Remark.

Phase is not unique:

$$e^{j\theta(\omega)} = e^{j(\theta(\omega) + 2\pi k)}, \quad k \in \mathbb{Z}.$$

Sometimes we force the **principal value**, $-\pi \leq \theta(\omega) < +\pi$.

2

DTFT Examples.

- $x[n] = \delta[n]$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\omega n} = \delta[0] e^{-j\omega \cdot 0} = 1.$$

- $x[n] = \alpha^n u[n]$, $|\alpha| < 1$:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} \alpha^n u[n] e^{-j\omega n} = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n \\ &= \frac{1}{1 - \alpha e^{-j\omega}} \quad \text{if } |\alpha e^{-j\omega}| < 1 \quad \left(|\alpha e^{-j\omega}| = |\alpha| \underbrace{|e^{-j\omega}|}_1 \right). \end{aligned}$$

Example (Rectangular Window Sequence).

$$\begin{aligned} x[n] &= \begin{cases} 1, & 0 \leq n \leq N-1 \\ 0, & \text{else} \end{cases} \\ &= u[n] - u[n-N]. \end{aligned}$$

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$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{N-1} e^{-j\omega n} \\ &= \frac{1 - (e^{-j\omega})^N}{1 - e^{-j\omega}} \\ &= \frac{e^{-j\omega \frac{N}{2}} (e^{+j\omega \frac{N}{2}} - e^{-j\omega \frac{N}{2}})}{e^{-j\omega \frac{1}{2}} (e^{+j\omega \frac{1}{2}} - e^{-j\omega \frac{1}{2}})} \\ &= \frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})} e^{-j\omega \frac{N-1}{2}}. \end{aligned}$$

2 Dirichlet Kernel

In signal processing, the Dirichlet kernel is often called the periodic sinc function:

$$\frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})}.$$

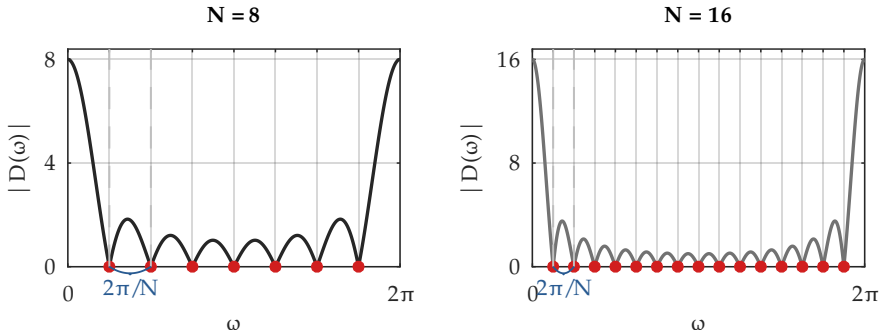
For small ω , $\sin(\frac{\omega}{2}) \approx \frac{\omega}{2}$, so

$$\frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})} \approx \frac{\sin(\omega \frac{N}{2})}{\frac{\omega}{2}} = N \operatorname{sinc}\left(\omega \frac{N}{2}\right), \quad \operatorname{sinc}(x) \triangleq \frac{\sin x}{x}.$$

Zeros occur when $\omega \frac{N}{2} = k\pi$, $k \neq 0$, i.e., $\omega = \frac{2\pi k}{N}$.



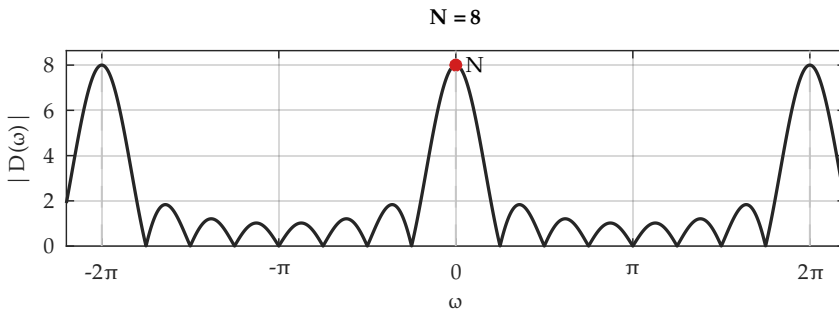
P. G. Lejeune Dirichlet



“more zeros” for larger N , faster oscillations.

4 But: also periodic (“periodic sinc”):

$$\left| \frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})} \right| \stackrel{?}{=} \left| \frac{\sin((\omega + 2\pi) \frac{N}{2})}{\sin(\frac{\omega + 2\pi}{2})} \right| = \left| \frac{\sin(\omega \frac{N}{2} + \pi N)}{\sin(\frac{\omega}{2} + \pi)} \right|.$$



$$|D(\omega)| = |D(\omega + 2\pi)|$$

Dealing with $0/0$ when $\omega = 0$:

$$X(e^{j\omega}) \Big|_{\omega=0} = \sum_{n=0}^{N-1} e^{-j\omega n} \Big|_{\omega=0} = N.$$

Limits: $\lim_{\omega \rightarrow 0} \frac{f(\omega)}{g(\omega)}$. For small ω , $\frac{\omega \frac{N}{2}}{\frac{\omega}{2}} = N$.

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5 3 DTFT Properties

3.1 Periodicity

Unlike the continuous-time Fourier transform, the DTFT is a periodic function of ω with period 2π .⁸

Property.

The DTFT is 2π -periodic: $X(e^{j(\omega+2\pi k)}) = X(e^{j\omega})$ for all $k \in \mathbb{Z}$.

Proof.

$$\begin{aligned} X(e^{j(\omega_0+2\pi k)}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega_0+2\pi k)n} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_0 n} \underbrace{e^{-j2\pi kn}}_{e^{-j2\pi kn}=1} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_0 n} = X(e^{j\omega_0}), \quad \text{for all values of } k. \quad \blacksquare \end{aligned}$$

⁷cf. Mitra Sec. 3.2.2, Basic Properties.

⁸cf. Mitra Sec. 3.2.1.

Remark*(addendum to orig lecture note).*

Why is $e^{-j2\pi kn} = 1$? Both k and n are integers, so $kn \in \mathbb{Z}$ and the angle $-2\pi kn$ is an integer multiple of 2π . Each full turn on the unit circle returns to 1:

$$e^{-j2\pi kn} = \cos(2\pi kn) - j \sin(2\pi kn) = 1.$$

3.2 Inverse transform**Inverse DTFT.**

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{+j\omega n} d\omega. \quad \text{(Mitra 3.14)}$$

Equations (3.10) and (3.14) form a DTFT pair. Write

$$\mathcal{F}\{x[n]\} = X(e^{j\omega}), \quad \mathcal{F}^{-1}\{X(e^{j\omega})\} = x[n],$$

and denote the DTFT pair by

$$x[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega}). \quad \text{(Mitra 3.15)}$$

⁹Mitra:

- It should be noted that even though the integration in Eq. (3.14) can be carried out over any interval of duration 2π , it is a common practice to choose the interval $[-\pi, \pi]$.
- The inverse discrete-time Fourier transform given by Eq. (3.14) can be interpreted as a linear combination of infinitesimally small complex exponential signals of the form $\frac{1}{2\pi} e^{j\omega n} d\omega$, weighted by the complex constant $X(e^{j\omega})$ over the normalized angular frequency range from $-\pi$ to π .
- Eq. (3.14) is referred to as the **synthesis equation**, because it synthesizes an arbitrary signal from its complex exponential components.

Proof.

$$\begin{aligned}
 x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{\ell=-\infty}^{\infty} x[\ell] e^{-j\omega\ell} \right) e^{j\omega n} d\omega \\
 &= \sum_{\ell=-\infty}^{\infty} x[\ell] \int_{-\pi}^{\pi} \frac{1}{2\pi} e^{j\omega(n-\ell)} d\omega \\
 &= \sum_{\ell=-\infty}^{\infty} x[\ell] \frac{\sin(\pi(n-\ell))}{\pi(n-\ell)} = \sum_{\ell=-\infty}^{\infty} x[\ell] \delta[n-\ell] \\
 &= x[n]. \quad \blacksquare
 \end{aligned}$$

6

3.3 Convergence

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Remark.

Exchanging the order of integration and summation (above) assumed convergence.

Convergence.

An infinite series of the form $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$ (3.10) may or may not converge. Let

$$X_K(e^{j\omega}) = \sum_{n=-K}^K x[n] e^{-j\omega n}. \quad (\text{Mitra 3.24})$$

denote the partial sum of the weighted complex exponentials in Eq. (3.10), and define

$$X(e^{j\omega}) = \lim_{K \rightarrow \infty} X_K(e^{j\omega})$$

when this limit exists (in the sense made precise below).

¹⁰cf. Mitra Sec. 3.2.4.

Def.

We say that $X_K(e^{j\omega})$ converges **uniformly** to $X(e^{j\omega})$ —stronger than point-wise convergence—if for every $\varepsilon > 0$, there exists an integer K such that for all integers $L \geq K$, the partial sum $X_L(e^{j\omega})$ of Eq. (3.24) satisfies

$$|X(e^{j\omega}) - X_L(e^{j\omega})| < \varepsilon \quad \text{for all } \omega.$$

Property.

If $x[n]$ is an **absolutely summable** sequence—that is, if Eq. (3.25) holds with

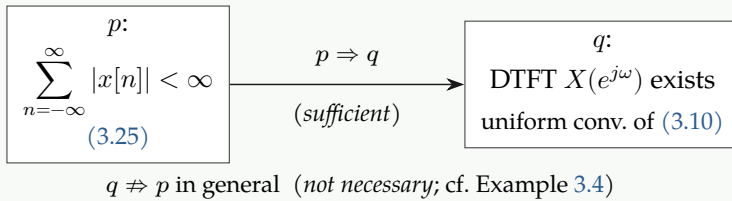
$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty. \quad (\text{Mittra 3.25})$$

then, by the triangle inequality and $|e^{-j\omega n}| = 1$,

$$|X(e^{j\omega})| = \left| \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x[n]| |e^{-j\omega n}| = \sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

for all ω , so the DTFT $X(e^{j\omega})$ of $x[n]$ exists for every ω .

Eq. (3.25) is thus a **sufficient condition** for existence of the DTFT $X(e^{j\omega})$:



Moreover, since $|x[n]e^{-j\omega n}| = |x[n]|$ is independent of ω , the Weierstrass M -test shows that the series of Eq. (3.10) converges **uniformly** for all ω .

Def.

We say that $X_K(e^{j\omega})$ converges in the **mean-square** to $X(e^{j\omega})$ if

$$\lim_{K \rightarrow \infty} \int_{-\pi}^{\pi} |X(e^{j\omega}) - X_K(e^{j\omega})|^2 d\omega = 0.$$

Property.

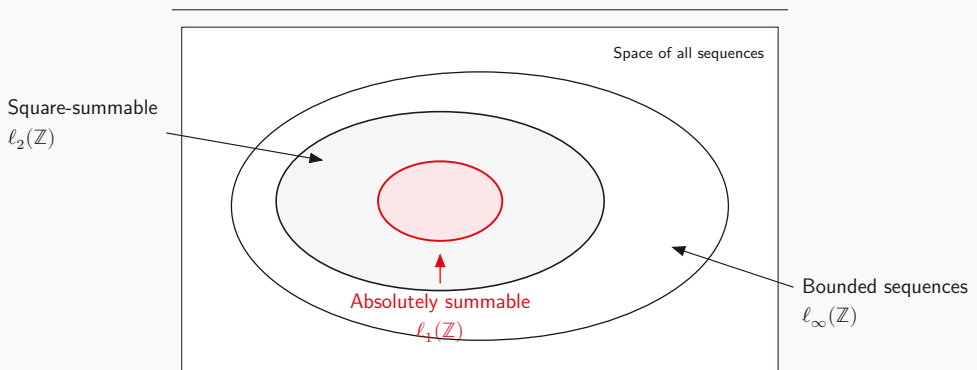
The DTFT will converge in the mean-square if the sequence $x[n]$ is **square-summable**:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty.$$

Property (Nesting of the $\ell_p(\mathbb{Z})$ spaces).

- Every absolutely summable sequence is also square summable.
- Every square-summable sequence is bounded.

$$\ell_1(\mathbb{Z}) \subset \ell_2(\mathbb{Z}) \subset \ell_\infty(\mathbb{Z}).$$



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3.4 An Example, and Generalized DTFTs

Example.

The sequence

$$x[n] = \begin{cases} 1/n, & n \geq 1 \\ 0, & n \leq 0 \end{cases}$$

is **not** absolutely summable, but **is** square summable (famous result by Euler):

$$\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^2 = \frac{\pi^2}{6}.$$

So $x[n] \in \ell_2(\mathbb{Z}) \setminus \ell_1(\mathbb{Z})$: absolute summability is **not necessary** for a sequence to admit a useful DTFT analysis in the mean-square sense (cf. the diagram in §3.3).

We can sometimes define DTFTs in cases where the DTFT does not actually converge (but still exists in a generalized sense).

Example.

$$x[n] = e^{j\omega_0 n}:$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} e^{j(\omega_0 - \omega)n} \quad \text{does not converge.}$$

But $x[n]$ still has a useful (generalized) DTFT:

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi k).$$

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$$X(e^{j\omega}) = 2\pi \underbrace{\delta(\omega - \omega_0)}_{\text{Dirac delta function}}, \quad -\pi \leq \omega \leq \pi, \quad -\pi \leq \omega_0 \leq \pi$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} 2\pi \delta(\omega - \omega_0) e^{j\omega n} d\omega = e^{j\omega_0 n}.$$

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 4.8
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 2.7

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 7: Linear Phase Filters

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Collected by Gewenxin Yu

Date September 15, 2026

1 Linear Phase Filters

Example (Delay System).

$$h[n] = \delta[n - M] \quad (\text{delay system})$$

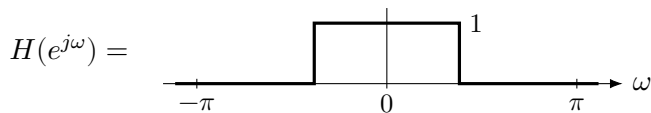
$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n - M] e^{-j\omega n} = e^{-j\omega M}.$$

$$|H(e^{j\omega})| = 1, \quad \theta(\omega) = -\omega M.$$

Here $\theta(\omega)$ is a **linear function of ω** (slope $-M$): phase grows in proportion to frequency, so **linear phase = constant delay**.

1.1 Why Linear Phase? Causal Approximation of Ideal Filters

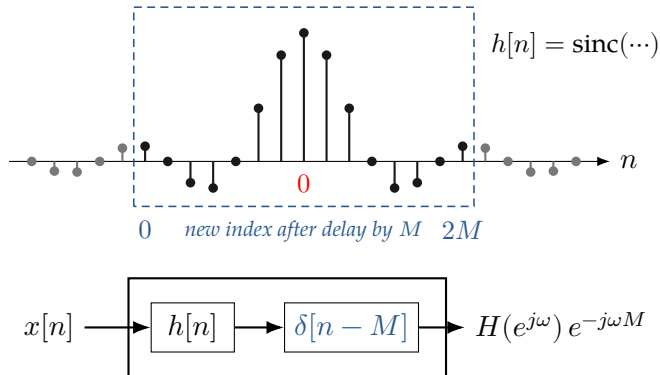
Linear phase is useful for **causal** systems. The ideal lowpass filter



has the non-causal, infinitely long impulse response

$$h[n] = \text{sinc}(\dots)$$

(symmetric about $n = 0$). *Truncate* it to a finite window $-M \leq n \leq M$ and **delay** it by M samples, so that the window starts at $n = 0$:



2 Generalized Linear Phase (GLP)

Def.

A system has **generalized linear phase (GLP)** if

$$H(e^{j\omega}) = \underbrace{R(e^{j\omega})}_{\text{real function}} e^{j(k_1 + \omega k_2)},$$

where $R(e^{j\omega})$ is real-valued (i.e., has phase 0 or π), and k_1, k_2 are constants.

GLP filters: **linear phase in the passband** (the sign flips of R only add jumps of π).

Property.

A **sufficient** condition for GLP for a real-valued, causal, length- N FIR filter is that

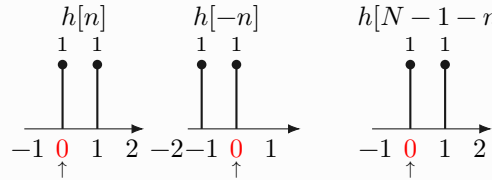
$$h[n] = \pm h[N - 1 - n] \quad \text{for } \forall n,$$

with $h[n] = 0 \quad \forall n < 0, n \geq N$.

If $h[0] \neq 0$ and $h[N - 1] \neq 0$, then the condition is also **necessary**.

Examples.

- $h[n] = \{1, 1\}$, $h[-n] = \{1, 1\}$, $h[N-1-n] = \{1, 1\}$ ✓



↑ marks $n = 0$ on each axis.

- $h[n] = \{1, -1\}$, $h[N-1-n] = \{-1, 1\}$ ✓
- $h[n] = \{1, -1, 1, -1\}$, $h[N-1-n] = \{-1, 1, -1, 1\}$ ✓ ($N = 4$, Type 4 antisymmetric)
- $h[n] = \{-1, 2, 1\}$, $h[0] = -h[2]$ but $h[1] = -h[1]$ fails ✗

Odd-length filters with antisymmetry cannot satisfy GLP unless the middle sample is 0.

with $h[n] = 0$ for $n \geq N$ and $n < 0$.

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2.1 Four Types of FIR GLP Filters

- Even symmetric or antisymmetric (+ or - in $h[n] = \pm h[N-1-n]$),
- N odd or even.

Notation: this handout vs. Mitra.

In this handout, N denotes the FIR filter **length**: the samples are $h[0], \dots, h[N-1]$, and the filter order is $N-1$. In Mitra Ch. 7, N denotes the filter **order**: the samples are $h[0], \dots, h[N]$, and the length is $N+1$. Hence

$$N_{\text{handout}} = N_{\text{Mitra}} + 1.$$

The Type 1–4 definitions are the same; only the meaning of the symbol N differs. For example, Mitra's Type 3 has even order N , hence odd length $N+1$, corresponding to §2.1.2 below.

2.1.1 N odd, symmetric and its proof (Mitra Type 1)

Assume $N = 2M + 1 \iff M = N - M - 1$, and $h[n] = h[N-1-n]$ for all n . Then

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=0}^{N-1} h[n] e^{-j\omega n} \\ &= \sum_{n=0}^{M-1} h[n] e^{-j\omega n} + \underbrace{h[M] e^{-j\omega M}}_{\text{middle point}} + \sum_{n=M+1}^{N-1} h[n] e^{-j\omega n} \\ &= e^{-j\omega M} \left(h[M] + \sum_{n=0}^{M-1} h[n] e^{-j\omega(n-M)} + \sum_{n=M+1}^{N-1} h[n] e^{-j\omega(n-M)} \right). \end{aligned}$$

In the **last sum**, let $n = N-1-k \iff k = N-1-n$:

$$\sum_{k=M-1}^0 h[N-1-k] e^{-j\omega(N-1-k-M)} = \sum_{k=0}^{M-1} h[N-1-k] e^{-j\omega(M-k)}.$$

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Renaming $\mathbf{k} \rightarrow \mathbf{n}$ and using $h[N - 1 - n] = h[n]$:

$$\begin{aligned}
 H(e^{j\omega}) &= e^{-j\omega M} \left[h[M] + \sum_{n=0}^{M-1} \left(\mathbf{h}[\mathbf{n}] e^{-j\omega(n-M)} \right. \right. \\
 &\quad \left. \left. + \underbrace{h[N - 1 - n]}_{\mathbf{h}[\mathbf{n}]} e^{+j\omega(\mathbf{n}-\mathbf{M})} \right) \right] \\
 &= \underbrace{e^{-j\omega M}}_{\text{linear phase}} \underbrace{\left[h[M] + \bullet \mathbf{2} \sum_{n=0}^{M-1} h[n] \mathbf{cos}(\omega[n - M]) \right]}_{\text{real function}}. \quad \blacksquare \quad (\text{Mittra 7.53})
 \end{aligned}$$

● **Remark**

(addendum to orig lecture note).

How to see $e^{-j\omega(n-M)} + e^{+j\omega(n-M)} = 2 \cos(\omega(n-M))$: this follows directly from **Euler's formula**. Let

$$\theta = \omega(n-M).$$

Then consider

$$e^{-j\theta} + e^{j\theta}.$$

From Euler's formula,

$$e^{j\theta} = \cos \theta + j \sin \theta,$$

and

$$e^{-j\theta} = \cos(-\theta) + j \sin(-\theta) = \cos \theta - j \sin \theta.$$

Therefore,

$$\begin{aligned} e^{-j\theta} + e^{j\theta} &= (\cos \theta - j \sin \theta) + (\cos \theta + j \sin \theta) \\ &= 2 \cos \theta. \end{aligned}$$

Substituting $\theta = \omega(n-M)$, we obtain

$$\boxed{e^{-j\omega(n-M)} + e^{j\omega(n-M)} = 2 \cos(\omega(n-M))}.$$

The intuition is that the two complex exponentials are complex conjugates:

$$\underbrace{\cos \theta - j \sin \theta}_{e^{-j\theta}} + \underbrace{\cos \theta + j \sin \theta}_{e^{j\theta}}.$$

Their imaginary parts cancel,

$$-j \sin \theta + j \sin \theta = 0,$$

while their real parts add:

$$\cos \theta + \cos \theta = 2 \cos \theta.$$

2.1.2 N odd, antisymmetric (odd symmetry) and its proof (Mitra Type 3)

Assume $N = 2M + 1$ is odd (so $M = \frac{N-1}{2} = N - M - 1$), $h[n] \in \mathbb{R}$ for all n , and $h[n] = -h[N - 1 - n]$ for all n (Type 3).

At the center tap, $h[M] = -h[N - 1 - M] = -h[M]$, hence $h[M] = 0$. Then

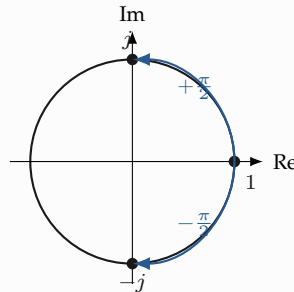
$$H(e^{j\omega}) = 2 e^{-j(\frac{\pi}{2} + \omega M)} \sum_{n=0}^{M-1} h[n] \sin(\omega[n - M]) \quad (\text{Recall: } \bullet e^{\pm j\frac{\pi}{2}} = \pm j). \quad (\text{Mitra 7.57})$$

● Remark

(addendum to orig lecture note).

Unit-circle picture for $e^{\pm j\pi/2} = \pm j$: each $e^{j\theta}$ is the point on the **unit circle** at angle θ counterclockwise from the positive real axis,

$$e^{j\theta} = \cos \theta + j \sin \theta.$$



At $\theta = +\frac{\pi}{2}$: $\cos(\frac{\pi}{2}) = 0$, $\sin(\frac{\pi}{2}) = 1$, so $e^{+j\pi/2} = j$. At $\theta = -\frac{\pi}{2}$: $e^{-j\pi/2} = -j$. Thus the two Recall values are quarter-turns from 1: **up** to $+j$, **down** to $-j$. In the formula above,

$$e^{-j(\frac{\pi}{2} + \omega M)} = \underbrace{e^{-j\pi/2}}_{\text{fixed quarter-turn} = -j} \underbrace{e^{-j\omega M}}_{\text{linear-phase factor varies with } \omega}.$$

Antisymmetry turns paired exponentials into sines and picks up this extra $\pm j$ factor—the “initial” phase offset $\pm \frac{\pi}{2}$ in the Note below—whereas symmetry ($h[n] = h[N - 1 - n]$) leaves a real cosine sum (phase 0 or π).

2.1.3 N even, antisymmetric (odd symmetry) and its proof (Mitra Type 4; from HW4)

Assume $N = 2M$ is even, $h[n] \in \mathbb{R}$ for all n , and $h[n] = -h[N - 1 - n]$ for all n (Type 4). There is no center tap at $n = \frac{N-1}{2} = M - \frac{1}{2}$; the impulse response is antisymmetric about the midpoint between samples $M - 1$ and M . Then

$$H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n] e^{-j\omega n} = \sum_{n=0}^{M-1} h[n] e^{-j\omega n} + \sum_{n=M}^{N-1} h[n] e^{-j\omega n}.$$

In the **last sum**, let $n = N - 1 - k$ (so $k = N - 1 - n$):

$$\sum_{k=0}^{M-1} h[N - 1 - k] e^{-j\omega(N-1-k)}.$$

5 Renaming **k** $\rightarrow$ **n** and using $h[N - 1 - n] = -h[n]$:

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=0}^{M-1} h[n] \left(e^{-j\omega n} - e^{-j\omega(N-1-n)} \right) \\ &= e^{-j\omega(M-\frac{1}{2})} \sum_{n=0}^{M-1} h[n] \left(e^{-j\omega(n-M+\frac{1}{2})} - e^{+j\omega(n-M+\frac{1}{2})} \right) \\ &= \underbrace{2 e^{-j(\frac{\pi}{2} + \omega(M-\frac{1}{2}))}}_{\text{linear phase}} \underbrace{\sum_{n=0}^{M-1} h[n] \sin(\omega[n - M + \frac{1}{2}])}_{\text{real function}}. \quad \blacksquare \quad (\text{Mitra 7.58}) \end{aligned}$$

The group delay is $\frac{N-1}{2} = M - \frac{1}{2}$ samples (a half-integer), as in the Type 1–4 table below; compare with §2.1.2 (Type 3), where odd N gives delay M and $\sin(\omega[n - M])$ instead of $\sin(\omega[n - M + \frac{1}{2}])$.

2.1.4 N even, symmetric and its proof (Mitra Type 2)

Assume $N = 2M$ is even, $h[n] \in \mathbb{R}$ for all n , and $h[n] = h[N - 1 - n]$ for all n (Type 2). The axis of symmetry is at $n = \frac{N-1}{2} = M - \frac{1}{2}$ (between samples $M - 1$

and M ; there is no center tap). Then

$$\begin{aligned}
 H(e^{j\omega}) &= \sum_{n=0}^{N-1} h[n] e^{-j\omega n} \\
 &= \sum_{n=0}^{M-1} h[n] e^{-j\omega n} + \sum_{n=M}^{N-1} h[n] e^{-j\omega n} \\
 &= e^{-j\omega(M-\frac{1}{2})} \left(\sum_{n=0}^{M-1} h[n] e^{-j\omega(n-M+\frac{1}{2})} + \sum_{n=M}^{N-1} h[n] e^{-j\omega(n-M+\frac{1}{2})} \right).
 \end{aligned}$$

In the **last sum**, let $n = N - 1 - k \iff k = N - 1 - n$:

$$\sum_{k=M-1}^0 h[N-1-k] e^{-j\omega(N-1-k-M)} = \sum_{k=0}^{M-1} h[N-1-k] e^{-j\omega(M-1-k)}.$$

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Renaming $\mathbf{k} \rightarrow \mathbf{n}$ and using $h[N-1-n] = h[n]$:

$$\begin{aligned}
 H(e^{j\omega}) &= e^{-j\omega(M-\frac{1}{2})} \left[\sum_{n=0}^{M-1} \left(\mathbf{h}[\mathbf{n}] e^{-j\omega(n-M+\frac{1}{2})} \right. \right. \\
 &\quad \left. \left. + \underbrace{h[N-1-n]}_{\mathbf{h}[\mathbf{n}]} e^{+j\omega(n-M+\frac{1}{2})} \right) \right] \\
 &= \underbrace{e^{-j\omega(M-\frac{1}{2})}}_{\text{linear phase}} \left[\underbrace{\bullet 2 \sum_{n=0}^{M-1} h[n] \cos(\omega[n-M+\frac{1}{2}])}_{\text{real function}} \right]. \quad \blacksquare \quad (\text{Mitra 7.56})
 \end{aligned}$$

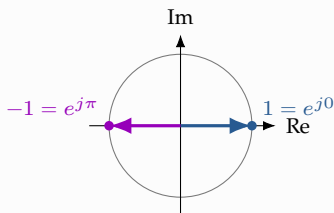
The group delay is $\frac{N-1}{2} = M - \frac{1}{2}$ samples. At $\omega = \pi$, each $\cos(\omega[n-M+\frac{1}{2}])$ is zero, so $H(e^{j\pi}) = 0$ (Type 2 cannot be a strict highpass); compare §2.1.1 (Type 1, (Mitra 7.53)), where length- N is odd and $\cos(\omega[n-M])$.

Note.

Recall from the beginning: $e^{-j\omega M}$ corresponds to a delay of M samples. GLP FIR filters are associated with a **delay of M samples** (larger $M \Rightarrow$ longer delay).

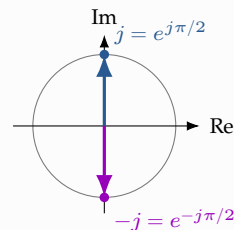
- Symmetric (+) $\Rightarrow$ “initial” phase of 0 or π .
- Antisymmetric (–) $\Rightarrow$ “initial” phase of $\pm \frac{\pi}{2} \Rightarrow$ good for: differentiation; Hilbert transform.

Symmetric (+): real axis



initial phase = 0 or π

Antisymmetric (–): imaginary axis



initial phase = $+\frac{\pi}{2}$ or $-\frac{\pi}{2}$

Remark

(addendum to orig lecture note).

Type	Symmetry	N (len.)	Forced zeros of $H(e^{j\omega})$
1	$h[n] = h[N-1-n]$	odd	none
2	$h[n] = h[N-1-n]$	even	$\omega = \pi$
3	$h[n] = -h[N-1-n]$ (odd symmetry)	odd	$\omega = 0$ and $\omega = \pi$
4	$h[n] = -h[N-1-n]$ (odd symmetry)	even	$\omega = 0$

- Type 2 cannot realize a highpass filter.
- Types 3/4 are antisymmetric (odd symmetry), have phase offset $\pm \frac{\pi}{2}$, and satisfy $H(e^{j0}) = 0$; they suit differentiators and Hilbert transformers.

5 3 Differentiators and Hilbert Transformers

3.1 Differentiator

$$\frac{dx(t)}{dt} \leftrightarrow j\Omega X(j\Omega) \quad \Longrightarrow \quad H(e^{j\omega}) \approx j\omega$$

(periodization and approximated appropriately).

$$H_{\text{DIF}}(e^{j\omega}) = j\omega, \quad 0 \leq |\omega| \leq \pi. \quad (\text{Mitra 7.100; 10.26})$$

GLP check (§2). With $R(e^{j\omega}) = \omega$ (real; phase 0 or π) and $k_1 = \frac{\pi}{2}$, $k_2 = 0$,

$$\begin{aligned}
 H_{\text{DIF}}(e^{j\omega}) &= R(e^{j\omega}) e^{j(k_1 + \omega k_2)} \\
 &= \omega e^{j\pi/2} && (\text{Recall: Euler } e^{j\theta} = \cos \theta + j \sin \theta, \theta = \frac{\pi}{2} \Rightarrow e^{j\pi/2} = j) \\
 &= j\omega, \quad \omega \in [-\pi, \pi].
 \end{aligned}$$

The fixed factor $e^{j\pi/2}$ is the same *initial phase* $\frac{\pi}{2}$ as in antisymmetric Types 3/4 (§2); here $k_2 = 0$, so there is no extra linear phase term $-\omega M$ beyond that offset.

3.2 Hilbert Transformer

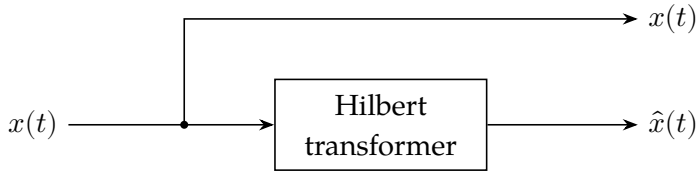
The ideal Hilbert transformer, also called a 90-degree phase shifter,¹¹ is characterized by a frequency response

$$H(j\Omega) = \begin{cases} -j, & \Omega \geq 0 \\ +j, & \Omega < 0 \end{cases}$$

(Here Ω is the continuous-time angular frequency, usually in rad/s.) **DTFT (ideal Hilbert transformer).** Frequency response $H(e^{j\omega})$:

$$H(e^{j\omega}) \approx \begin{cases} -j, & 0 \leq \omega < \pi \\ +j, & -\pi \leq \omega < 0 \end{cases} \quad (\text{Mitra 10.24})$$

(Here ω is the discrete-time normalized angular frequency, in rad/sample.)



Generation of an analytic signal using a Hilbert transformer. (Mitra Figure 1.3, Sec. 1.2.3)

Form the complex signal

$$z(t) = x(t) + j\hat{x}(t) \quad (\text{Mitra 1.8})$$

where $\hat{x}(t) = x(t) \otimes h(t)$:

$$Z(j\Omega) = \begin{cases} X(j\Omega) + X(j\Omega), & \Omega \geq 0 \\ X(j\Omega) - X(j\Omega), & \Omega < 0 \end{cases} = \begin{cases} 2X(j\Omega), & \Omega \geq 0 \\ 0, & \Omega < 0 \end{cases} \quad (\text{Mitra 1.9})$$

¹¹Ideal Hilbert transform filtering has $|H(\Omega)| = 1$ for all Ω (**allpass** magnitude): phase $-\frac{\pi}{2}$ at each positive frequency and $+\frac{\pi}{2}$ at each negative frequency. This is why Mitra calls it a **90-degree phase shifter**. Contrast §5.1 (differentiator), where $|H| \propto |\Omega|$. Cf. J. O. Smith, "The Analytic Signal and Hilbert Transform Filters" (§5.3.4).

¹²The signals $x(t)$ and $\hat{x}(t)$ are called, respectively, the **in-phase** and **quadrature** components of $z(t)$.

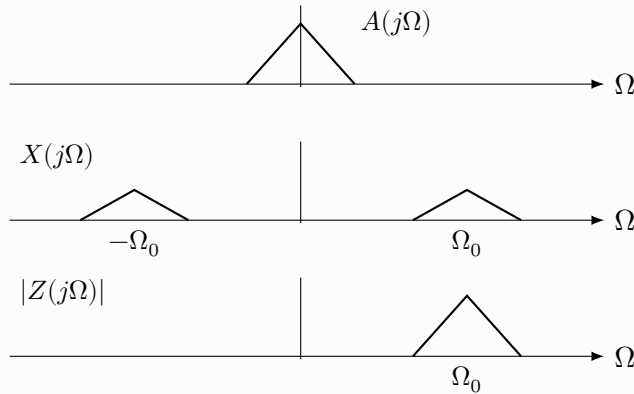
⇒ created a complex-valued signal with a **single-sided** Fourier transform (the *analytic signal*)¹³.

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Example (Amplitude Modulation)

(Mitra Sec. 1.2.4).

Let $a(t)$ be a lowpass signal with spectrum $A(j\Omega)$, and $x(t) = a(t) \cos(\Omega_0 t + \phi)$.



Assuming the bandwidth of $a(t)$ is smaller than Ω_0 , form the analytic signal from §5.2 ($z = x + j\hat{x}$, (1.8)):

$$z(t) = x(t) + j\hat{x}(t), \quad \hat{x}(t) = \mathcal{H}\{x(t)\}.$$

The Hilbert transformer applies $-j$ on $\Omega \geq 0$ (§5.2); for a **narrowband** real AM waveform, the envelope varies slowly compared with the carrier, so the phase shift acts on $\cos(\Omega_0 t + \phi)$ while $a(t)$ is treated as fixed:

$$\hat{x}(t) = \mathcal{H}\{a(t) \cos(\Omega_0 t + \phi)\} \approx a(t) \mathcal{H}\{\cos(\Omega_0 t + \phi)\} = a(t) \sin(\Omega_0 t + \phi).$$

Carrier check ($\theta = \Omega_0 t + \phi$): $\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$; the $+\Omega_0$ and $-\Omega_0$ terms pick up

¹³Why form $z = x + j\hat{x}$? The analytic signal can be written $z(t) = A(t)e^{j\psi(t)}$, with $|z(t)| = A(t)$ the **instantaneous envelope** and $\psi(t)$ the instantaneous phase (so $f(t) = \frac{1}{2\pi} \frac{d\psi}{dt}$ is instantaneous frequency). That is the usual motivation for envelope detection, AM demodulation, and related bandpass processing once $\hat{x} = \mathcal{H}\{x\}$ is available. Cf. accepted answer to “Meaning of Hilbert Transform”: <https://dsp.stackexchange.com/questions/25845/meaning-of-hilbert-transform>.

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$-j$ and $+j$ (§5.2), so $\mathcal{H}\{\cos \theta\} = \sin \theta$. With $\bullet e^{j\theta} = \cos \theta + j \sin \theta$,

$$z(t) = a(t) \cos \theta + j a(t) \sin \theta = a(t) e^{j(\Omega_0 t + \phi)}.$$

Hence

$$\Rightarrow \begin{cases} |z(t)| = |a(t)| & \text{(envelope detection),} \\ a(t) = z(t) e^{-j(\Omega_0 t + \phi)} & \text{(demodulation).} \end{cases}$$

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4 Phase Delay and Group Delay

Phase Delay.

$$\tau_p(\omega_0) \triangleq -\frac{\theta(\omega_0)}{\omega_0}. \quad (\text{Mitra 4.95})$$

Indeed, for input $e^{j\omega_0 n}$,

$$\begin{aligned} e^{j\omega_0 n} \rightarrow \boxed{h[n]} &\rightarrow |H(e^{j\omega_0})| e^{j(\omega_0 n + \theta(\omega_0))} = |H(e^{j\omega_0})| e^{j\omega_0 \left(n - \left(\frac{-\theta(\omega_0)}{\omega_0}\right)\right)} \\ &= |H(e^{j\omega_0})| e^{j\omega_0 (n - \tau_p(\omega_0))}. \end{aligned}$$

Group Delay.

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega}. \quad (\text{Mitra 4.96})$$

- **Constant group delay:** relative phases of different frequency components will be maintained.
- **GLP:** constant $\tau_g(\omega)$ for almost all ω (except at the jumps of π).
- **Non-constant phase or group delay:** distortion.¹⁴

¹⁴Example: HW4 Problem 5(b)—filters (2), (4), (6), and (7) from that set's LTI list. On a symmetric test input, $\tau_g(\omega)$ varies with ω , frequency components misalign, and the pulse shape is warped (GLP filters (1) and (5) preserve symmetry).

8 5 All-Pass Filters

All-Pass Filter.

$|H(e^{j\omega})| = 1$, while $\theta(\omega)$ is typically **not** linear. (All-pass filters can compensate nonlinear phase from other systems; that is why we sometimes intentionally add all-pass (nonlinear-phase) stages for **phase equalization**, without changing magnitude.)

Example (LDE).

$$\sum_{k=0}^M d_k y[n-k] = \sum_{k=0}^M d_{M-k}^* x[n-k].$$

If the frequency response is convergent,

$$H(e^{j\omega}) = \frac{\sum_{k=0}^M d_{M-k}^* e^{-j\omega k}}{\sum_{k=0}^M d_k e^{-j\omega k}}, \quad |H(e^{j\omega})| = 1.$$

Remark

(addendum to orig lecture note).

Why $|H| = 1$ in general: substituting $m = M - k$ in the numerator,

$$\sum_{k=0}^M d_{M-k}^* e^{-j\omega k} = e^{-j\omega M} \sum_{m=0}^M d_m^* e^{+j\omega m} = e^{-j\omega M} \left(\sum_{m=0}^M d_m e^{-j\omega m} \right)^*,$$

so the numerator and denominator have equal magnitude. The example below checks this directly.

Example.

$$d_k = \{1, \frac{1}{2}\}_{\uparrow}$$

$$y[n] + \frac{1}{2} y[n-1] = \frac{1}{2} x[n] + x[n-1] \iff y[n] = -\frac{1}{2} y[n-1] + \frac{1}{2} x[n] + x[n-1].$$

Now we prove that this is an all-pass filter:

$$H(e^{j\omega}) = \frac{\frac{1}{2} + e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}} = \frac{\frac{1}{2} + \cos \omega - j \sin \omega}{1 + \frac{1}{2} \cos \omega - \frac{j}{2} \sin \omega},$$

We take the square of the magnitude:

$$\begin{aligned} |H(e^{j\omega})|^2 &= \frac{\left(\frac{1}{2} + \cos \omega\right)^2 + \sin^2 \omega}{\left(1 + \frac{1}{2} \cos \omega\right)^2 + \left(\frac{1}{2} \sin \omega\right)^2} \quad 15 \\ &= \frac{\frac{1}{4} + \cos \omega + \overbrace{\cos^2 \omega + \sin^2 \omega}^{=1}}{1 + \cos \omega + \frac{1}{4} \cos^2 \omega + \frac{1}{4} \sin^2 \omega} = \frac{\frac{5}{4} + \cos \omega}{\frac{5}{4} + \cos \omega} = 1. \end{aligned}$$

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Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 5.2
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 6

¹⁸Use two facts:

- $\left|\frac{A}{B}\right|^2 = \frac{|A|^2}{|B|^2}$
- For $z = x + jy$: $|z|^2 = x^2 + y^2$

Expanding with Euler's formula $e^{-j\omega} = \cos \omega - j \sin \omega$:

- Numerator: $\frac{1}{2} + \cos \omega - j \sin \omega \Rightarrow |\cdot|^2 = \left(\frac{1}{2} + \cos \omega\right)^2 + \sin^2 \omega$
- Denominator: $1 + \frac{1}{2} \cos \omega - \frac{j}{2} \sin \omega \Rightarrow |\cdot|^2 = \left(1 + \frac{1}{2} \cos \omega\right)^2 + \left(\frac{1}{2} \sin \omega\right)^2$
- The minus signs vanish when squared
- Dividing the two gives the first step: $|H(e^{j\omega})|^2 = \frac{\left(\frac{1}{2} + \cos \omega\right)^2 + \sin^2 \omega}{\left(1 + \frac{1}{2} \cos \omega\right)^2 + \left(\frac{1}{2} \sin \omega\right)^2}$

Appendix

- A. DSP Foundations (Euler, Analytic Signals, Norms)

A DSP Foundations (Euler, Analytic Signals, Norms)

Euler's formula and variants

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (\text{Euler's formula}).$$

Identities used in GLP proofs, Fourier pairs, and Hilbert/analytic-signal material:

$$\begin{aligned} e^{j\theta} + e^{-j\theta} &= 2 \cos \theta, & \cos \theta &= \frac{1}{2}(e^{j\theta} + e^{-j\theta}), \\ e^{j\theta} - e^{-j\theta} &= 2j \sin \theta, & \sin \theta &= \frac{1}{2j}(e^{j\theta} - e^{-j\theta}), \\ e^{j\pi/2} &= j, & e^{-j\pi/2} &= -j. \end{aligned}$$

Analytic signal

Analytic signal.

A signal whose spectrum has **no negative-frequency components** is called an **analytic signal**.

Continuous time. For real $x(t)$ with Fourier transform $X(j\Omega)$, form

$$z(t) = x(t) + j \hat{x}(t), \quad \hat{x}(t) = \mathcal{H}\{x(t)\},$$

where $\mathcal{H}\{\cdot\}$ is an ideal Hilbert transformer ($\mp j$ on $\Omega \gtrless 0$). Then $z(t)$ is analytic: $Z(j\Omega) = 0$ for $\Omega < 0$ (positive frequencies retained; cf. Lecture 7).

Discrete time. Similarly $z[n] = x[n] + j \hat{x}[n]$ with $\hat{x}[n] = x[n] \otimes h_{\text{HT}}[n]$ and ideal $H_{\text{HT}}(e^{j\omega}) = \mp j$ on $[0, \pi) / (-\pi, 0)$ gives $Z(e^{j\omega}) = 0$ on $-\pi \leq \omega < 0$.

In-phase / quadrature: x is the **in-phase** and $\hat{x}$ the **quadrature** component of z . Often $|z(t)|$ is the **instantaneous envelope** and $\arg z(t)$ the instantaneous phase (narrowband / AM settings).

Common norms and energy

Discrete-time sequences $x[n]$ (when the sums converge):

$$\|x\|_1 = \sum_{n=-\infty}^{\infty} |x[n]|, \quad \|x\|_2 = \sqrt{\sum_{n=-\infty}^{\infty} |x[n]|^2}, \quad \|x\|_{\infty} = \sup_{n \in \mathbb{Z}} |x[n]|.$$

Continuous-time signals $x(t)$ (when the integrals converge):

$$\|x\|_1 = \int_{-\infty}^{\infty} |x(t)| dt, \quad \|x\|_2 = \sqrt{\int_{-\infty}^{\infty} |x(t)|^2 dt}, \quad \|x\|_{\infty} = \sup_{t \in \mathbb{R}} |x(t)|.$$

Parseval (DTFT).

When the DTFT exists in the usual sense,

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = \|X\|_2^2 \quad (\text{energy in time} = \text{energy in frequency}).$$

Ref.: J. O. Smith, “The Analytic Signal and Hilbert Transform Filters” (§5.3.4), https://ccrma.stanford.edu/~jos/st/Analytic_Signals_Hilbert_Transform.html.